REPORT

Title

Decision Tree Classification Using Python

Aim

To implement a **Decision Tree classification algorithm** using Python and predict the class of new data based on given input features.

Problem Statement

Decision Tree is a supervised machine learning algorithm used for classification and prediction. It represents a decision-making process in the form of a tree. Internal nodes represent conditions, branches represent possible outcomes, and leaf nodes represent the final class.

The objective is to develop a Decision Tree model that classifies students as Pass or Fail based on their study hours and attendance.

Requirements

- Python 3.x
- Python IDLE / VS Code / Jupyter Notebook
- Scikit-learn library
- Basic knowledge of Python and machine learning

Procedure

1. Create a dataset containing study hours and attendance.
2. Assign Pass or Fail as the target class.
3. Divide the dataset into training and testing data.
4. Create a `DecisionTreeClassifier`.

5. Train the model using the training dataset.
6. Test the trained model using the testing dataset.
7. Compare the predicted values with the actual values.
8. Calculate the classification accuracy.
9. Provide new student data to the trained model.
10. Display the predicted result.

Program

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

X = [
    [1, 50],
    [2, 55],
    [3, 60],
    [4, 65],
    [5, 70],
    [6, 75],
    [7, 80],
    [8, 85],
    [9, 90],
    [10, 95]
]

y = [0, 0, 0, 0, 1, 1, 1, 1, 1, 1]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = DecisionTreeClassifier(random_state=42)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Actual Values   :", y_test)
print("Predicted Values:", y_pred)

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy * 100, "%")

new_student = [[6, 78]]
prediction = model.predict(new_student)

if prediction[0] == 1:
    print("New Student: Pass")
```

```
else:
    print("New Student: Fail")
```

Sample Output

```
Actual Values   : [1 0]
Predicted Values: [1 0]
Accuracy: 100.0 %
New Student: Pass
```

Conclusion

The Decision Tree classification algorithm was successfully implemented using Python. The model was trained using study hours and attendance and successfully predicted whether a new student would pass or fail. The algorithm provides an easy-to-understand tree-based approach for classification problems.